

Dr Abdulzeid Yen Anafo

Lecturer in Mathematics and Statistics · University of Mines and Technology, Tarkwa, Ghana

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ACADEMIC QUALIFICATIONS

PhD, Mathematics (Mathematical Statistics) 2019–2022

University of Mines and Technology, Ghana

- *Dissertation:* Development of the Modified Extended Chen and Equilibrium Renewal Burr XII Distributions Using the Burr–Hatke Differential Equation and Equilibrium Renewal Model, with Applications to Health and Climate Data
- *Supervisors:* Prof. Suleman Nasiru and Prof. Lewis Brew

MSc, Mathematical Sciences 2018–2019

African Institute for Mathematical Sciences (AIMS), Ghana

BSc, Mathematical Sciences 2013–2017

University for Development Studies, Ghana

GRANT AWARDS

Sub-Principal Investigator — Global Health Emerging Scholars Program (GHES) 2024–2025

National Institutes of Health (NIH)

- “Estimating Climatic and Human Health Impact of Open Burning of Scrap Tires as Fuel for Singeing in Ghanaian Meat Production Processes”
- Led design and implementation of field research protocols on the environmental and public health impacts of open scrap-tyre burning in livestock processing
- Supervised data collection teams across sites, ensuring research ethics, data quality standards and CAPI procedures
- Conducted quantitative analysis of air quality exposure, health outcomes and climatic implications
- Coordinated engagement with government agencies, EPA units, public health authorities, abattoir management and community leaders
- Prepared technical reports, policy briefs and dissemination materials; supported budget monitoring and NIH research-administration compliance

RESEARCH APPOINTMENTS

Post-Doctoral Fellow

Apr 2024 – Dec 2025

Yale University, USA

- Interdisciplinary research on global health equity, focused on environmental pollution and health
- Designed and implemented context-specific research projects with international partners and mentors

Post-Doctoral Fellow

Apr 2023 – Mar 2024

West Africa Mathematical Modelling Project, Ghana (partner: Northwestern University)

- Modelled seasonal malaria chemoprevention intervention strategies in the Northern Region of Ghana
- Produced District Health Survey reports on children under five with malaria

Visiting Research Scholar

Jun – Oct 2023

Institute for Global Health, Feinberg School of Medicine, Northwestern University, Chicago, USA · Malaria Modelling Program (PI: Dr Jaline Gerardin)

- Intensive training on EMOD; collaborated on malaria intervention modelling for Seasonal Malaria Chemoprevention in Ghana
- Aligned Ghana-specific epidemiological data with global modelling frameworks

TEACHING

Lecturer

Current

University of Mines and Technology, Tarkwa

- Mathematical Analysis · Probability and Statistics · Probability Theory · Time Series Analysis

Adjunct Faculty

Mar 2024

Ashesi University, Ghana

- Taught differential equations and applied mathematics to master's students
- Supervised and assessed postgraduates in the MSc Mechatronics programme

Technical Instructor — Epidemiological MODelling (EMOD)

2024, 2025

Faculty Enrichment Program, University of Health and Allied Sciences, Ghana

- Delivered remote instruction on EMOD to faculty and researchers in applied malaria modelling
- Designed sessions on simulating malaria transmission dynamics and evaluating interventions
- Mentored participants building customised disease models

PROFESSIONAL EXPERIENCE

Data Scientist (remote consultant)

Apr – Jun 2025

NylaBank, Ghana

- Built and optimised machine learning models for customer segmentation
- Exploratory data analysis and Power BI / Metabase dashboards on transaction and login data

Health Statistician

Sep 2019 – Dec 2022

Ministry of Health, Ghana

- Analysed and reported on allied health professionals nationally; mapped them in R
- Contributed to an integrated payment system for the Allied Health Professions Council
- Built a data collection application for the AHPC and an algorithm to extract default professional renewals from its management information system

PUBLICATIONS

1. **Anafo, A. Y.**, Ocloo, S. and Nasiru, S. (2025). **Modified alpha power transformed Topp-Leone family of distributions with applications and regression.** *Scientific African*, 30, e02993.
2. **Anafo, A. Y.**, Bosson-Amedenu, S., Gmayinaam, V. U., Enock, A., Ocloo, S. and Acquah, J. (2025). **Evaluating malaria burden in children under-five and intervention outcomes in Tarkwa-Nsuaem municipality.** *BMC Infectious Diseases*, 25(1), 294.
3. Bosson-Amedenu, S., **Anafo, A. Y.**, Ouerfelli, A., Ouerfelli, N. and Ouerfelli, N. (2024). **Examining Cesarean section rates in Ghana's 10 regions over a decade: a comprehensive national investigation.** *BioMed Research International*, 3774435.
4. **Anafo, A. Y.**, Ocloo, S. K. and Nasiru, S. (2024). **New weighted Burr XII distribution: statistical properties, applications and regression.** *International Journal of Mathematics and Mathematical Sciences*, 4098771.
5. Acquah, J., Eyiah-Bediako, F. and **Anafo, A. Y.** (2024). **Covariate level stratification assessment of mortalities due to competing risks and breast cancer in Ghana.** *Scientific African*.
6. **Anafo, A. Y.**, Selasi, O. and Napari, H. M. (2024). **Logit Gudermannian distribution: properties, regression and applications.** *Journal of Statistical Modelling: Theory and Applications*.
7. Acquah, J., Odoi, B. and **Anafo, A. Y.** (2023). **An extension of the Chen distribution: properties, simulation study and applications to data.** *Asian Journal of Probability and Statistics*.
8. **Anafo, A. Y.**, Brew, L. and Nasiru, S. (2023). **The modified extended Chen distribution: properties and applications to rainfall data.** *Applied Mathematics and Information Sciences*.
9. **Anafo, A. Y.**, Brew, L. and Nasiru, S. (2021). **The equilibrium renewal Burr XII distribution: properties and applications.** *Asian Journal of Probability and Statistics*.

10. Ampadu, C. and **Anafo, A. Y.** (2019). **The new rT–X family of distributions: some properties with applications.** *Earthline Journal of Mathematical Sciences*. doi:10.34198/ejms.2219.409432
11. Ampadu, C. and **Anafo, A. Y.** (2019). **Maximum likelihood estimation in an alpha power transformation of the Ampadu-type.** *Annals of Biostatistics & Biometric Applications*. doi:10.33552/ABBA.2019.02.000541
12. **Anafo, A. Y.** and Ampadu, C. (2019). **The generalized Ampadu-G family of distributions: properties and applications.** *Earthline Journal of Mathematical Sciences*, 4(1), 139–167. doi:10.34198/ejms.4120.139167

ONGOING RESEARCH

Predicting depression among men who have sex with men in Ghana using machine learning algorithms Accepted

PLOS Mental Health · github.com/Zeidyen/msm-depression-ml

Characterisation of air pollutants from singeing sites in Ghana Completed

github.com/Zeidyen/Ghana-singeing-air-quality

Assessing the impact of seasonal malaria chemoprevention on older age groups in Northern Ghana Completed

github.com/Zeidyen/SMC-Malaria-Ghana-Project

EDITORIAL AND PEER REVIEW

2024– Associate Editor, *Frontiers in Applied Mathematics and Statistics*

2023– Reviewer, Hindawi

2023– Reviewer, *Scientific African*

2023– Reviewer, *PLOS ONE*

2023– Reviewer, *Heliyon*

INVITED AND CONFERENCE PRESENTATIONS

Scenario Modelling for Strategic Decision-Making: From Mathematical Frameworks to Policy Impact 2025

Applied Malaria Modelling Network — SME Working Group Meeting, Dakar, Senegal · Oral

Modelling the Impact of Expanding Seasonal Malaria Chemoprevention for the Older Age Group and Five Cycles on Malaria Incidence in the Northern Region of Ghana 2024

MIM Society 8th Pan-African Conference, Kigali, Rwanda · Poster · with J. Gerardin, Y. Afrane and J. Aheto

Impact Analysis of Perennial Malaria Chemoprevention on Malaria Incidence in Northern Region of Ghana

2024

2nd Annual Symposia, Conakry, Guinea · Oral · with J. Gerardin, Y. Afrane and J. Aheto

Modelling the Epidemiological Impact of Extending SMC Towards a Broader Age Group

2023

1st Annual Symposia, West Africa Mathematical Modelling Capacity Development Project, Accra, Ghana · Oral · with J. Gerardin, Y. Afrane and J. Aheto

PROFESSIONAL TRAINING

2024 Geospatial Modelling — West Africa Mathematical Modelling Capacity Development, Accra

2022 Data Warehousing Workshop — Ministry of Health, Accra

2022 Health Information Management System — World Bank, Accra

2020 Machine Learning Professional Course — Blossom Academy, Accra (*certificate*)

AWARDS, HONOURS AND MEMBERSHIPS

2023– Member, American Society of Tropical Medicine and Hygiene

2023– Member, Applied Malaria Modelling Network

2020 Ghana National Petroleum Corporation Postgraduate Scholarship

2018 African Institute for Mathematical Sciences Scholarship

TECHNICAL SKILLS

Statistics R, Python, SPSS — statistical analysis and simulation

Modelling EMOD — disease and intervention models

Reporting Power BI, Metabase, Microsoft Office

Languages English — fluent (read, written, spoken)

REFEREES

Available on request.

Contact by email. Telephone and postal details are not published on this page.